

//Q44: Write a program to find the sum of the series: $1 + \frac{3}{4} + \frac{5}{6} + \frac{7}{8} + \dots$ up to n terms.

```
/*
Sample Test Cases:
Input 1:
3
Output 1:
Approximate sum: 3.3

Input 2:
5
Output 2:
Approximate sum: 4.4

*/
#include <stdio.h>

int main() {
float n,num,den,sum=0;
printf("enter a num ");
if(scanf("%f",&n)!=1){
printf("invalid input");}

if(n>=1) sum=sum+1;

for(int i=2;i<=n;i++){
num=2*i-1;
den=2*i;
sum=sum+(num/den);}

printf("%f",sum);

return 0;
}
```

//Q45: Write a program to find the sum of the series: $\frac{2}{3} + \frac{4}{7} + \frac{6}{11} + \frac{8}{15} + \dots$ up to n terms.

```
/*
Sample Test Cases:
Input 1:
3
Output 1:
Approximate sum: 1.56

Input 2:
5
Output 2:
Approximate sum: 2.22

*/
#include <stdio.h>
int main() {
float n,num,den,sum=0;
printf("enter a num ");
if(scanf("%f",&n)!=1){
printf("invalid input");}

for(int i=1;i<=n;i++){
num=i*2;
den=i*4-1;
sum=sum+(num/den);}

printf("%f",sum);

return 0;
}
```

Q46: Write a program to print the following pattern:

Click to collapse the range.

***** */

/*

Sample Test Cases:

Input 1:

Output 1:

*/

#include <stdio.h>

int main() {

int n;

printf("enter a num ");

if(scanf("%d",&n)!=1){

printf("invalid input");}

for(int i=1;i<=n;i++){

for(int i=1;i<=n;i++){

printf("*");}

printf("\n");}

return 0;

}

```
/*  
**  
***  
****  
***** */
```

```
#include <stdio.h>
```

```
int main() {  
    int n;  
    printf("enter the num ");  
    if(scanf("%d",&n)!=1){  
        printf("invalid input");}
```

```
    for(int i=1;i<=n;i++){  
        for(int j=1;j<=i;j++){  
            printf("*");  
        }  
        printf("\n");}
```

```
    return 0;  
}
```

```
1 #include <stdio.h>
2 int main () {
3     int n;
4     printf("enter the num ");
5     if(scanf("%d",&n)!=1){
6         printf("invalid input");}
7
8
9     for(int i=1;i<=n;i++){
10         for(int j=1;j<=i;j++){
11             printf("%d",j);}
12
13         printf("\n");}
14
15
16
17
18     return 0;
19 }
```

/*Q48: Write a program to print the following pattern:

```
1
12
123
1234
12345 */
```

/*

Sample Test Cases:

Input 1:

Output 1:

```
1
12
123
1234
12345
```

*/

```
#include <stdio.h>
```

```
int main() {
```

```
int n;
```

```
printf("enter the num ");
```

```
if(scanf("%d",&n)!=1){
```

```
printf("invalid input");}
```

```
for(int i=1;i<=n;i++){
```

```
for(int j=1;j<=i;j++){
```

```
printf("%d",j);}
```

```
printf("\n");}
```

```
return 0;
```

```
}
```

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1 /*Q49: Write a program to print the following pattern:

2 5

3 45

4 345

5 2345

6 12345 */

7 /*

8 Sample Test Cases:

9 Input 1:

1 Output 1:

2 5

3 45

4 345

5 2345

6 12345

7 */

8 #include <stdio.h>

1 int main() {

2 int n;

3 printf("enter the num ");

4 if(scanf("%d",&n)!=1){

5 printf("invalid input");}

6 for(int i=1;i<=n;i++){

7 for(int j=n+1-i;j<=n;j++){

8 printf("%d",j);}

9 printf("\n");}

10 return 0;

/*Q51: Write a program to print the following pattern:

```
    5
   45
  345
 2345
12345 */
```

/*

Sample Test Cases:

Input 1:

Output 1:

```
    5
   45
  345
 2345
12345
```

*/

```
#include <stdio.h>
```

```
int main() {
```

```
int n;
```

```
printf("enter the num ");
```

```
if(scanf("%d",&n)!=1){
```

```
printf("invalid input");}
```

```
for(int i=1;i<=n;i++){
```

```
for(int j=1;j<=n+1-i;j++){
```

```
printf(" ");}
```

```
for(int k=n+1-i;k<=n;k++){
```

```
printf("%d",k);}
```

```
printf("\n");
```

```
}
```

```
    return 0;
```

```
}
```

Output 1:

```
*****
```

```
*****
```

```
***
```

```
**
```

```
*
```

Input 2:

Output 2:

Note: Spaces indicate indentation.

```
*/
```

```
#include <stdio.h>
```

```
✓ int main() {  
  int n;  
  printf("enter the num ");  
✓ if(scanf("%d",&n)!=1){  
  printf("invalid input");}
```

```
✓ for(int i=1;i<=n;i++){  
✓ for(int j=1;j<=i;j++){  
  printf(" ");}  
✓ for(int k=1;k<=n+1-i;k++){  
  printf("*");}  
  printf("\n");}
```

```
    return 0;
```

```
},
```

```
/*
 *
 **
 ***
 ****
 *****
 ****
 ***
 **
 * */
```

```
#include <stdio.h>
int main () {
int n;
printf("enter the num ");
if(scanf("%d",&n)!=1){
printf("invalid input");}

for(int i=1;i<=n;i++){
for(int j=1;j<=i;j++){
printf("*");}
printf("\n");}
for(int i=2;i<=n;i++){
for(int j=1;j<=n+1-i;j++){
printf("*");}
printf("\n");}

return 0;
}
```

Pattern with layers of stars as shown.

```
*/  
#include <stdio.h>  
✓ int main () {  
    int n;  
    printf("enter the num ");  
✓ if(scanf("%d",&n)!=1){  
    printf("invalid input");}  
  
    int nst=1;  
    int ml=(n+1)/2;  
    int nsp=ml-1;  
✓ for(int i=1;i<=n;i++){  
✓ for(int a=1;a<=nsp;a++){  
    printf(" ");}  
✓ for(int b=1;b<=nst;b++){  
    printf("*");}  
    printf("\n");  
✓ if(ml>i){  
    nsp--;  
    nst+=2;}  
✓ else{  
    nsp++;  
    nst-=2;}}  
  
    return 0;  
}
```

```
*/  
#include <stdio.h>  
int main() {  
    int n;  
    printf("enter the num ");  
    if(scanf("%d",&n)!=1){  
        printf("invalid input");}  
  
    printf("*\n\n");  
    for(int i=1;i<=3;i++){  
        printf("*\n");}  
    printf("\n");  
    for(int i=1;i<=5;i++){  
        printf("*\n");}  
    printf("\n");  
    for(int i=1;i<=3;i++){  
        printf("*\n");}  
    printf("\n");  
    printf("*\n\n");  
    return 0;  
}
```